

# TechMart — How to Run

## Environment Setup and Deployment Guide (Jakarta EE 10 / Payara Server)

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### 1. Prerequisites Overview

Before deploying TechMart, four things need to be installed and configured on your machine, in this order:

1. **JDK 17**
2. **MySQL Server 8.x**
3. **Payara Server 6**
4. **Payara resources** — JDBC connection pool, JDBC resource, JMS connection factory, and JMS queue

Once these are in place, the application is deployed as a single EAR file and accessed at <http://localhost:8080/techmart/pos>.

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### 2. Install JDK 17

TechMart's `pom.xml` targets Java 17 (`maven.compiler.source/target`), so an older or newer JDK can cause build or deployment issues.

1. Download **JDK 17** (Temurin/Adoptium build is recommended) from <https://adoptium.net/temurin/releases/?version=17>
  2. Install it, then confirm it's on your PATH:  
  
`java -version`  
  
Expected output should start with `openjdk version "17....`
  3. If you have multiple JDKs installed, make sure `JAVA_HOME` points at the JDK 17 install directory, since both Maven and Payara will use whatever `JAVA_HOME` resolves to.
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### 3. Install MySQL Server

TechMart persists everything (products, customers, orders, inventory, notifications) in MySQL 8.

1. Install **MySQL Server 8.x** (via MySQL Installer on Windows, `brew install mysql` on macOS, or your Linux distro's package manager).
2. Start the MySQL service and make sure you can log in:

```
mysql -u root -p
```

3. Run the seed script to create the techmart database, schema, and sample data. Use **database-seed.sql** (not the original `Seed_SQL`) — it has boosted inventory quantities so load testing doesn't fail on stock exhaustion:

```
mysql -u root -p < database-seed.sql
```

4. Confirm the tables were created:

```
USE techmart;  
SHOW TABLES;  
SELECT COUNT(*) FROM products;
```

5. Note the credentials and connection details (host, port — default 3306, database name techmart, username, password) — you'll need them when configuring the Payara connection pool in Section 5.
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### 4. Install and Start Payara Server

1. Download **Payara Server 6** (Community edition) from <https://www.payara.fish/downloads/>

2. Unzip it to a location of your choice, e.g. C:\payara6 or ~/payara6.
3. Start the domain using asadmin, Payara's command-line admin tool, found in payara6/bin/:

```
cd payara6/bin
./asadmin start-domain
```

On Windows, use asadmin.bat start-domain from Command Prompt or PowerShell.

4. Confirm it started successfully — you should see:

```
Domain domain1 started.
```

5. Open the Payara Admin Console in a browser at:

```
http://localhost:4848
```

This is where the resources in the next section are configured.

To stop the server later: `./asadmin stop-domain`

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## 5. Configure the JDBC Connection Pool and Resource

The application looks up its datasource via the JNDI name `jdbc/TechMartDataSource` (as declared in `techmart-ejb/src/main/resources/META-INF/persistence.xml`) and expects a **MySQL** driver.

### 5.1 Create the Connection Pool

In the Admin Console: **Resources** → **JDBC** → **JDBC Connection Pools** → **New**

| Field                | Value                             |
|----------------------|-----------------------------------|
| Pool Name            | TechMartPool                      |
| Resource Type        | javax.sql.DataSource              |
| Datasource Classname | com.mysql.cj.jdbc.MySQLDataSource |
| Driver Classname     | com.mysql.cj.jdbc.Driver          |

On the next page, under **Additional Properties**, add:

| Name         | Value     |
|--------------|-----------|
| serverName   | localhost |
| portNumber   | 3306      |
| databaseName | techmart  |

| Name     | Value                      |
|----------|----------------------------|
| user     | <i>your MySQL username</i> |
| password | <i>your MySQL password</i> |

Recommended **Pool Settings** for load testing:

| Field                         | Value       |
|-------------------------------|-------------|
| Initial and Minimum Pool Size | 8           |
| Maximum Pool Size             | 200         |
| Pool Resize Quantity          | 30          |
| Idle Timeout                  | 300 seconds |
| Max Wait Time                 | 60000 ms    |

Click **Ping** to verify the pool connects successfully before saving.

## 5.2 Create the JDBC Resource

**Resources → JDBC → JDBC Resources → New**

| Field     | Value                   |
|-----------|-------------------------|
| JNDI Name | jdbc/TechMartDataSource |
| Pool Name | TechMartPool            |

The JNDI name must match **exactly** — jdbc/TechMartDataSource. If it's registered under a different name (e.g. jdbc/TechMartDS), the EAR will fail to inject the persistence context at deploy time.

## 6. Configure JMS Resources (Connection Factory and Queue)

PlaceOrderServlet sends a message to a queue after checkout, and OrderProcessingMDB consumes it asynchronously. Two JMS resources are required.

### 6.1 JMS Connection Factory

**Resources → JMS Resources → Connection Factories → New**

| Field         | Value                         |
|---------------|-------------------------------|
| JNDI Name     | jms/TechmartConnectionFactory |
| Resource Type | jakarta.jms.ConnectionFactory |

### 6.2 JMS Destination Resource (Queue)

First create the physical destination: **Resources → JMS Resources → Destination Resources → New**

| Field                     | Value                  |
|---------------------------|------------------------|
| JNDI Name                 | jms/TechmartOrderQueue |
| Physical Destination Name | TechmartOrderQueue     |
| Resource Type             | jakarta.jms.Queue      |

Both JNDI names must match the `@Resource(lookup = "...")` values in `PlaceOrderServlet.java` and the `destinationLookup` value in `OrderProcessingMDB.java` exactly, or deployment/message delivery will fail silently or with lookup errors.

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## 7. Build and Deploy the Application

From the project root, build the multi-module Maven project:

```
mvn clean install
```

This produces `techmart-ear/target/techmart-ear.ear`.

### 7.1 Deploy via Admin Console

1. Go to **Applications** → **Deploy** in the Admin Console.
2. Choose the generated `techmart-ear.ear` file.
3. Leave the context root as `/techmart` (already set in `techmart-ear/pom.xml`).
4. Click **OK** to deploy.

### 7.2 Or deploy via `asadmin`

```
./asadmin deploy /path/to/techmart-ear/target/techmart-ear.ear
```

Check the server log (`payara6/glassfish/domains/domain1/logs/server.log`) if deployment fails — this is the most reliable place to see the real JNDI/driver error rather than the generic browser error page.

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## 8. Access the Application

Once deployed, open a browser and go to:

```
http://localhost:8080/techmart/pos
```

This loads the POS terminal — product catalogue, customer selector, and shopping cart. From here you can:

- Add products to the cart and place an order (`/techmart/placeOrder`)
- View recent orders and notifications at `/techmart/transactions`
- View live system metrics at `/techmart/metrics`

